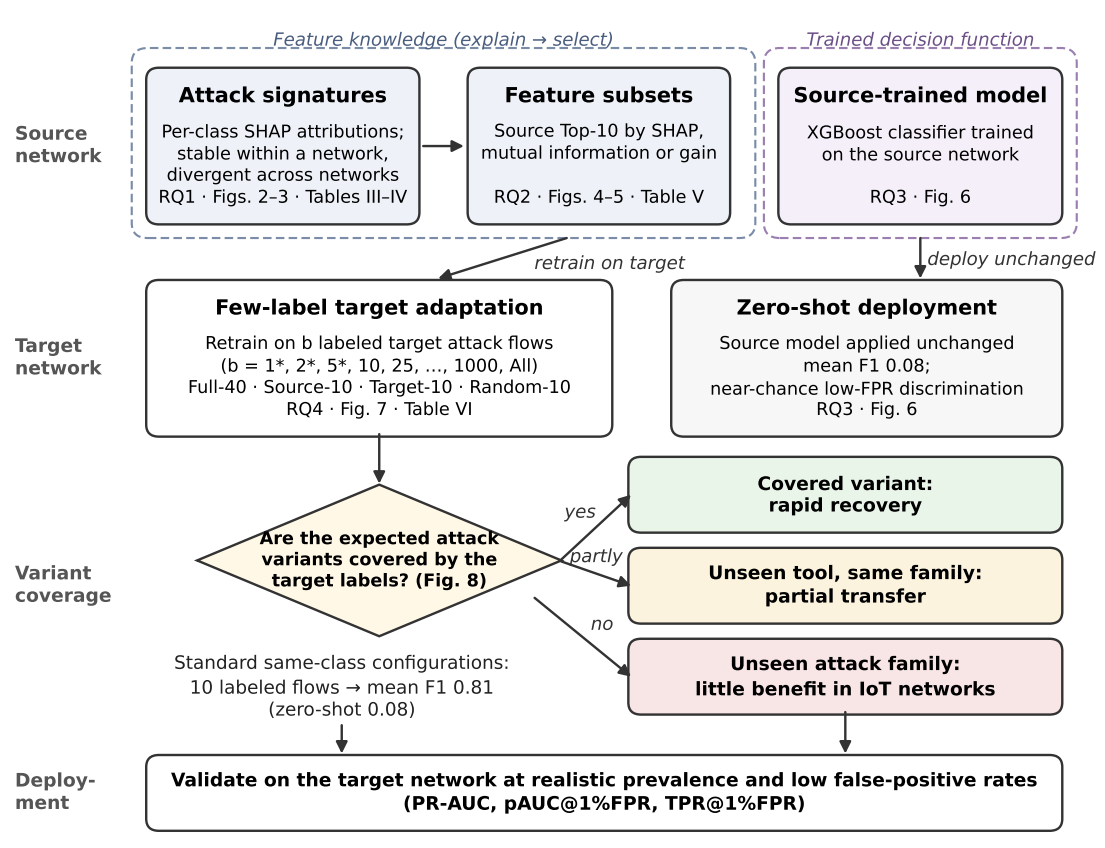
Feature knowledge (explain → select)
Trained decision function
Source network
Attack signatures
Per-class SHAP attributions; stable within a network, divergent across networks
RQ1 · Figs. 2–3 · Tables III–IV
Feature subsets
Source Top-10 by SHAP, mutual information or gain
RQ2 · Figs. 4–5 · Table V
Source-trained model
XGBoost classifier trained on the source network
RQ3 · Fig. 6
retrain on target
deploy unchanged
Target network
Few-label target adaptation
Retrain on b labeled target attack flows (b = 1*, 2*, 5*, 10, 25, ..., 1000, All)
Full-40 · Source-10 · Target-10 · Random-10
RQ4 · Fig. 7 · Table VI
Zero-shot deployment
Source model applied unchanged
mean F1 0.08;
near-chance low-FPR discrimination
RQ3 · Fig. 6
Variant coverage
Are the expected attack variants covered by the target labels? (Fig. 8)
yes
partly
no
Covered variant: rapid recovery
Unseen tool, same family: partial transfer
Unseen attack family: little benefit in IoT networks
Standard same-class configurations:
10 labeled flows → mean F1 0.81
(zero-shot 0.08)
Deploy-ment
Validate on the target network at realistic prevalence and low false-positive rates (PR-AUC, pAUC@1%FPR, TPR@1%FPR)